DS-A3 | Regression Baseline and Diagnostics

Dataset: scikit-learn Diabetes dataset (442 observations, 10 predictors) https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_diabetes?

Fixed split: 80/20, random_state=42

Local CSV SHA-256: f16718c1e6602b419193b9a023dbe278ae7f85ff343158813d7040a9f7512dec

1. Dataset and method

The scikit-learn Diabetes dataset contains 442 observations and 10 predictors. The local CSV was frozen and identified by a SHA-256 hash. A fixed 80/20 train–test split used random_state=42; the test set was not used for model selection. A training-mean baseline, ordinary linear regression, and Ridge regression were compared. Ridge's alpha was selected by five-fold cross-validation using training data only.

Main result: Ridge achieved test RMSE 53.777 and R^2 0.454, only a marginal improvement over linear regression (RMSE 53.853, R^2 0.453). Both regression models outperformed the naive-mean baseline. The model remains a baseline rather than a deployment-ready system because diagnostics and slice analysis show issues that need further investigation.

Model	MAE	RMSE	R^2
Naive mean	64.006	73.222	–0.012
Linear regression	42.794	53.853	0.453
Ridge (α=1)	42.812	53.777	0.454

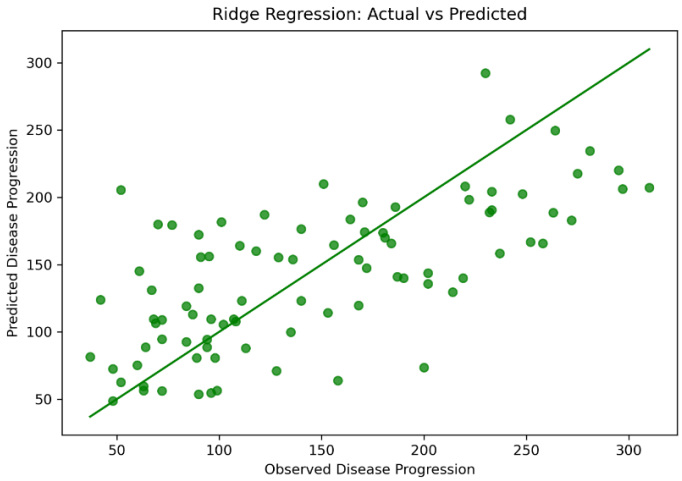


Figure 1. Ridge regression: actual versus predicted

Predictions rise with observed disease progression, but the points are widely scattered around the reference line. The model captures a useful trend while leaving substantial individual prediction error.

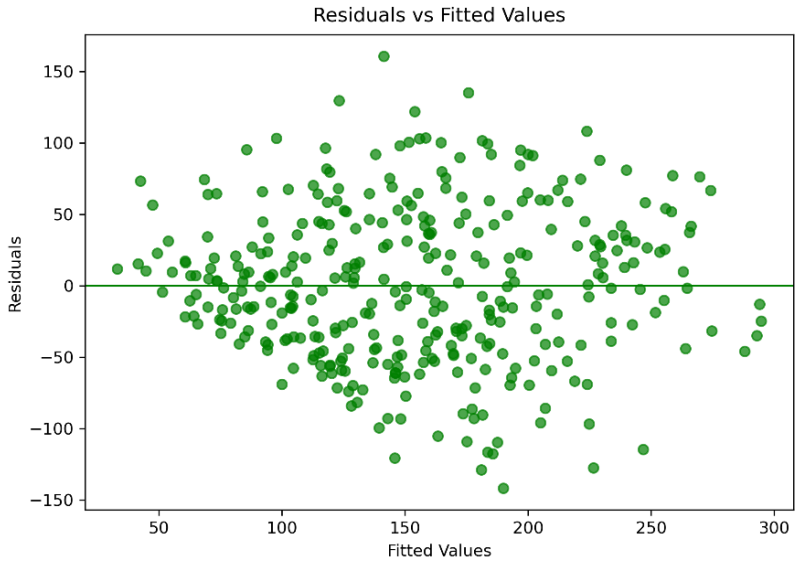


Figure 2. Residuals versus fitted values

Residuals fall on both sides of zero, but their spread varies across fitted values and several large errors remain. This supports checking non-constant variance and investigating large residuals.

2. Diagnostic Evidence

Residual and influence checks help assess model assumptions and identify observations that may affect the fitted relationship. They complement, but do not replace, held-out performance metrics.

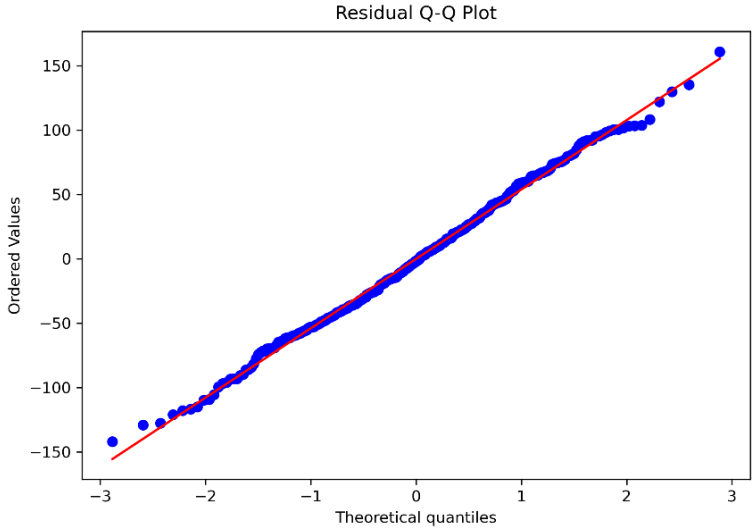


Figure 3. Residual Q–Q plot

Most points follow the reference line reasonably closely, suggesting approximate normality. Deviations at the tails show that the most extreme residuals do not follow the line perfectly.

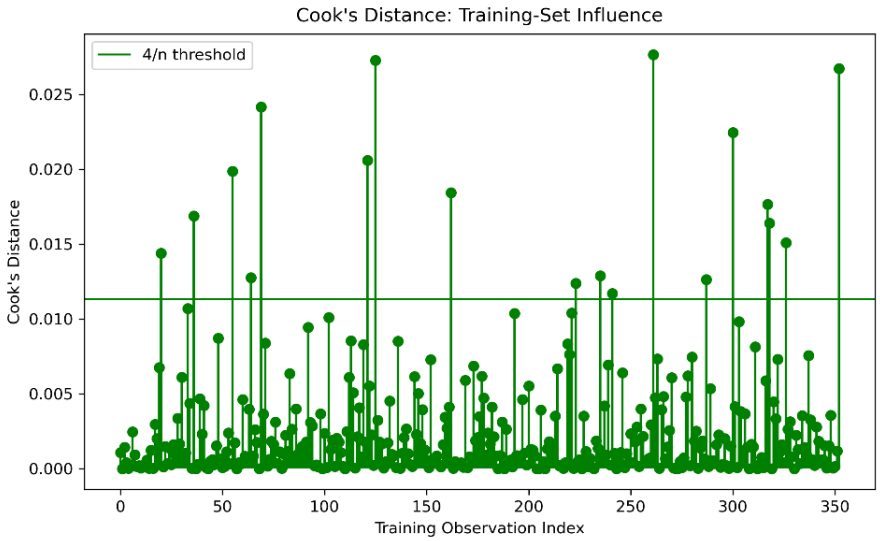


Figure 4. Cook's distance

Eighteen observations exceed the 4/n screening threshold; the maximum Cook's distance is 0.0277. Review these cases, but treat the threshold as a screening guide rather than an automatic reason for removal.

3. Generalization and Subgroup Performance

Learning curves show how performance changes with training-set size, while slice analysis checks whether errors are similar across a meaningful subgroup.

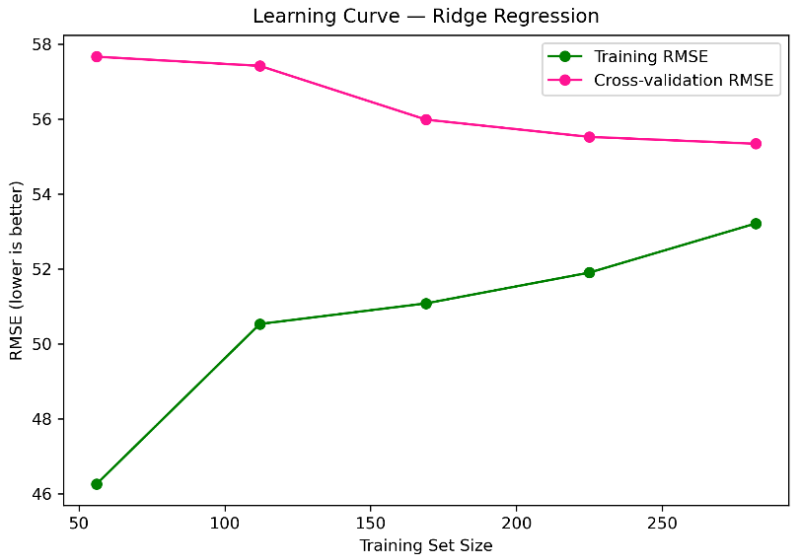


Figure 5. Ridge regression learning curve

Cross-validation RMSE falls as training size increases, while training RMSE rises. The gap narrows, but validation error remains higher, suggesting more representative data or better features may help.

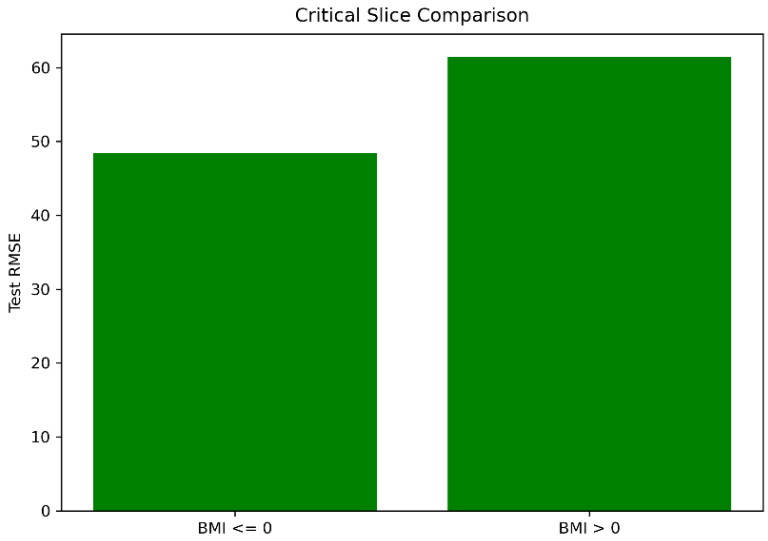


Figure 6. Critical-slice comparison by centered BMI

The BMI-above-mean slice has higher RMSE (about 61.47) than the at-or-below-mean slice (about 48.42). This shows that average performance can hide subgroup differences.

4. Model Comparison and Final Assessment

Ridge has the lowest RMSE (53.777) and highest R^2 (0.454), but its gain over ordinary linear regression is tiny. Both regression models outperform the naive-mean baseline on this fixed test set.

Table 1. Model comparison on the frozen test set

Model Comparison on Frozen Test Set

Model	MAE	RMSE	R2
Naive mean	64.006	73.222	-0.012
Linear regression	42.794	53.853	0.453
Ridge	42.812	53.777	0.454

Breusch–Pagan p=0.0345 indicates evidence of non-constant residual variance. Shapiro–Wilk p=0.6398 does not provide evidence against normality; Cook's-distance screening identifies cases for inspection.

Figure 8. Regression diagnostics summary

Regression Diagnostics Summary

Diagnostic	Value
Breusch-Pagan p-value	0.0345
Shapiro-Wilk p-value	0.6398
Max Cook's distance	0.0277
Influential points (Cook's > 4/n)	18.0

5. Conclusion

Ridge is the best of the three models by RMSE and R^2, but the gain over linear regression is negligible. The Breusch–Pagan result suggests non-constant variance; influential observations and higher error in the BMI-above-mean slice should be reviewed. Before deployment, conduct external validation, examine error cases and subgroup performance, and document limitations. Observed associations should not be interpreted as causal effects.

Reproducibility: retain the executable notebook, frozen dataset, environment/dependency record, and result CSV files with this report.